

Global Comparison Report

Monte Carlo Strategy Paired-Difference Analytics

GENERATED	MONTE CARLO RUNS	STRATEGIES TESTED
10 May 2026 13:56	30	2
METHOD	UNIT	REPORT TYPE
Discrete-Event Simulation (DES)	Haemodialysis Centre	Comparative Performance Analysis

Abstract

This report presents the results of a Monte Carlo discrete-event simulation study evaluating scheduling strategies for a haemodialysis unit. Stochastic variability in patient volume, machine availability, session duration, and nursing resources is modelled across 30 independent runs per strategy. Distributional summaries, tail-risk metrics (p90 / p99), and head-to-head strategy comparisons are provided to support operational decision-making.

Keywords: Monte Carlo | Discrete-Event Simulation | Haemodialysis | Scheduling | Tail-Risk

1. Simulation Setup

The parameters below fully specify the stochastic environment sampled during each Monte Carlo iteration. All stochastic quantities are drawn independently per run using a seeded Mersenne Twister PRNG.

TEMPORAL BOUNDS

Parameter	Value	Unit	Description
Shift Duration	360	min	Hard wall -- 6 h total shift
Nominal Session Duration	210 - 240	min	Prescribed per-patient session (~4 h target)
Min Viable Session	180	min	Sessions shorter than this are marked failed (3 h)
Machine Cooldown	60	min	Mandatory inter-session machine rest period

RESOURCE RANGES (uniform draw per iteration)

Parameter	Value	Unit	Description
Total Machines	15 - 20	units	Physical dialysis stations available
Patient Volume	15 - 20	patients	Demand sampled each shift
Nursing Staff	2 - 4	nurses	Concurrent supervision capacity
Session Duration	210 - 240	min	Per-patient dialysis time
Machine Ready Delay	0 - 90	min	Post-cooldown preparation lag

STOCHASTIC EVENT GENERATORS

Parameter	Value	Unit	Description
Patient Arrival Offset	Uniform[0, 60]	min	Arrival jitter relative to shift start
Setup Duration	Uniform[10, 20]	min	Machine preparation time per patient

PROBABILITY METRICS

Parameter	Value	Unit	Description
Machine Defect Probability	15.0%	--	Independent Bernoulli draw per machine per session

EXPERIMENT PARAMETERS

Parameter	Value	Unit	Description
Monte Carlo Iterations	30	runs	Independent replications per strategy

2. Global Statistical Summary

Per-strategy distributional metrics (mean +/- SD, p50, p90, p99) aggregated across the full Monte Carlo suite. The p90 and p99 columns serve as tail-risk indicators; lower is better.

FIFO

Metric	Mean +/- SD	p50	p90	p99
Mean Wait Time (min)	28.8 +/- 12.3	28.2	47.1	51.0

Max Wait Time (min)	52.6 +/- 19.2	49.0	80.2	91.6
Avg Session Time (min)	224.9 +/- 2.5	224.8	227.3	230.7

FIXED

Metric	Mean +/- SD	p50	p90	p99
Mean Wait Time (min)	58.4 +/- 18.2	54.1	80.3	88.1
Max Wait Time (min)	109.6 +/- 25.0	104.0	138.8	152.3
Avg Session Time (min)	221.0 +/- 4.9	221.8	227.6	228.6

3. Comparative Analytics

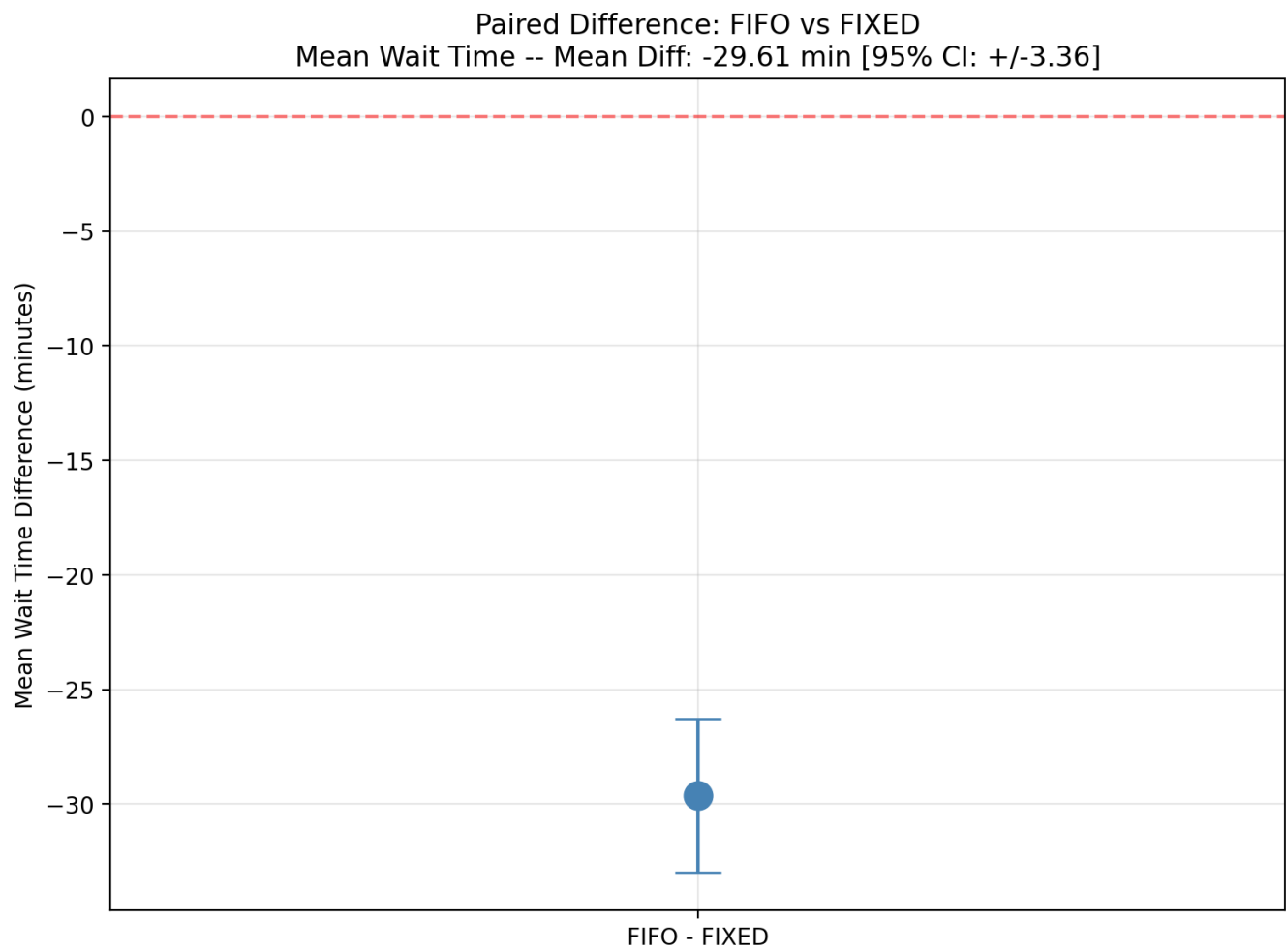


Figure 1. Paired-difference analysis of mean wait time across strategies.

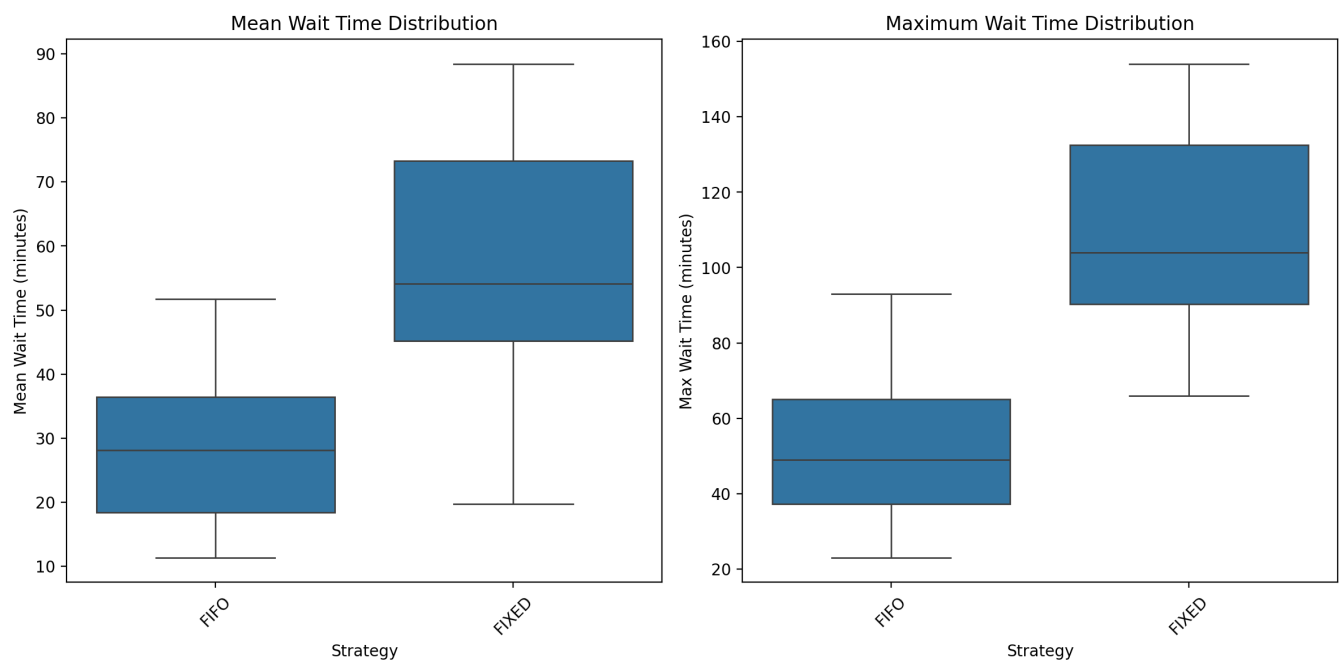


Figure 2. Wait-time distribution comparison across strategies.

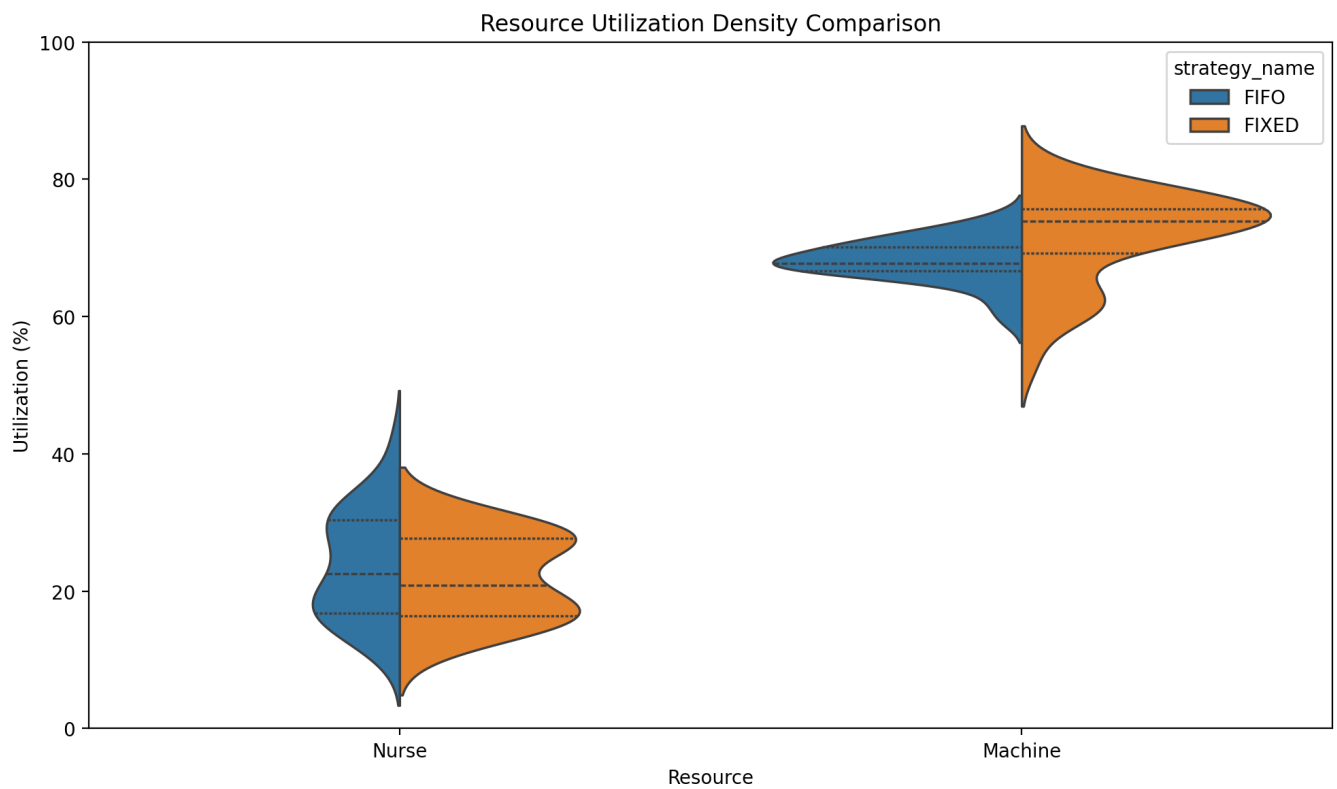


Figure 3. Utilisation violin plots across strategies.

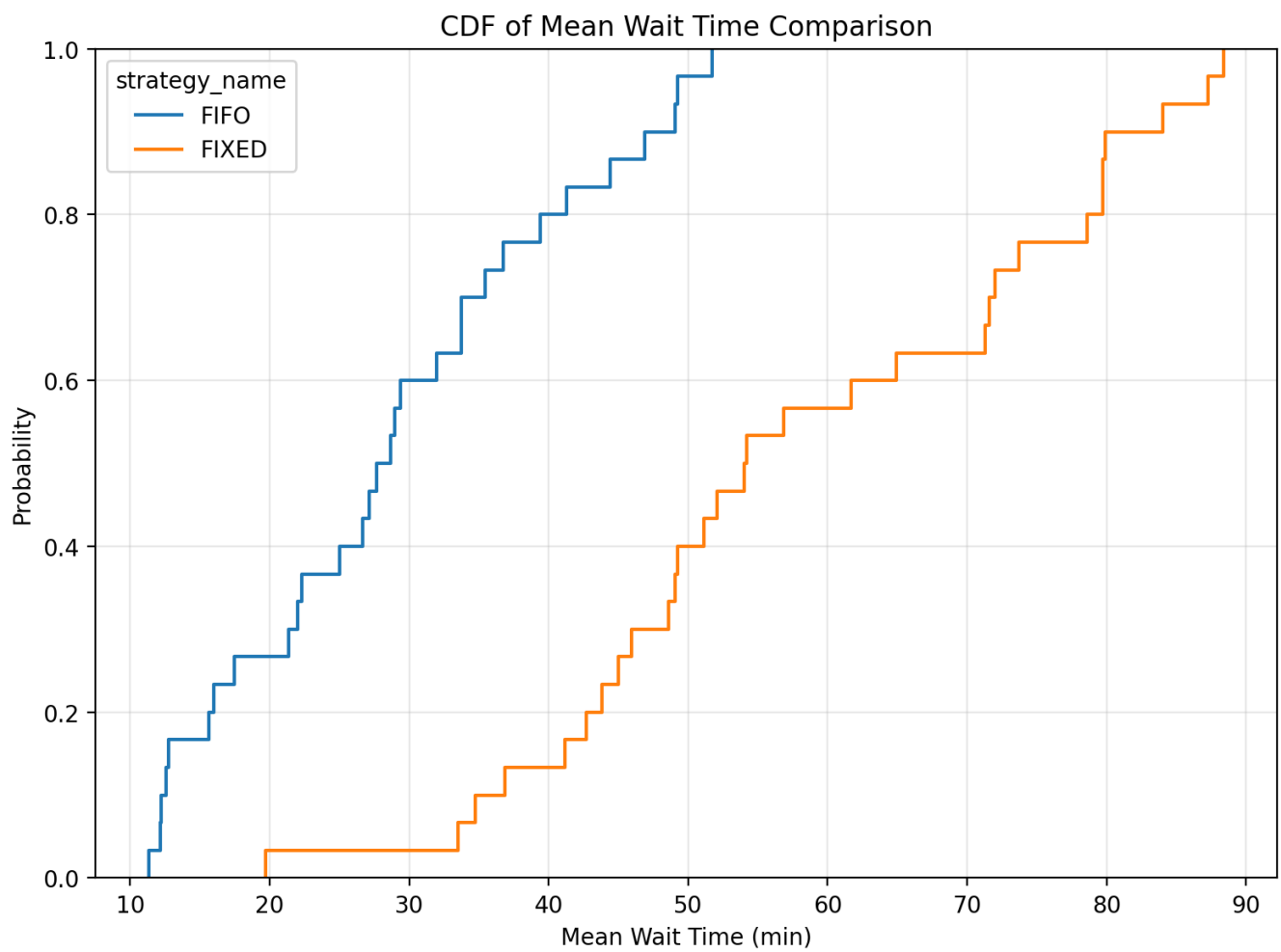


Figure 4. Empirical CDF of mean wait time -- all strategies.

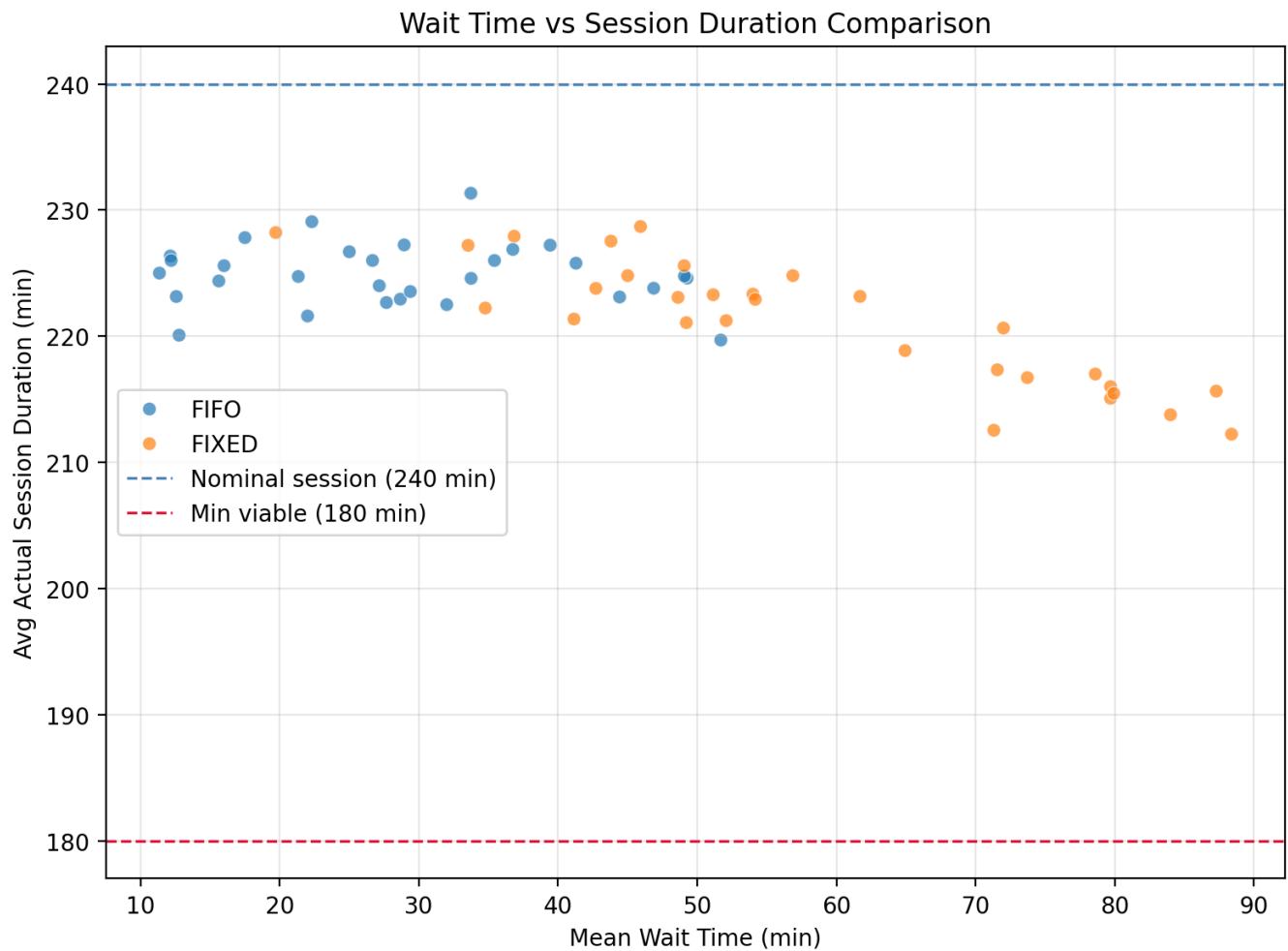


Figure 5. Wait time vs. avg session duration -- all strategies.

4. Edge-Case Stress Tests

Performance under extreme, deterministic boundary conditions designed to expose failure modes not visible in the central Monte Carlo distribution.

Scenario: Burst Arrival

[FAIL]" if failed else "[PASS] Strategy: FIFO | Wait (mean / max): 66.5 / 139.0 min | Avg session: 220 min | Truncated: 4 | Failed patient: 4

[FAIL]" if failed else "[PASS] Strategy: FIXED | Wait (mean / max): 66.5 / 139.0 min | Avg session: 220 min | Truncated: 4 | Failed patient: 4

Scenario: Maintenance Disaster

[FAIL]" if failed else "[PASS] Strategy: FIFO | Wait (mean / max): 0.0 / 0.0 min | Avg session: 215 min | Truncated: 0 | Failed patient: 0

[FAIL]" if failed else "[PASS] Strategy: FIXED | Wait (mean / max): 0.0 / 0.0 min | Avg session: 228 min | Truncated: 0 | Failed patient: 0

Scenario: Marathon Shift

[FAIL]" if failed else "[PASS] Strategy: FIFO | Wait (mean / max): 0.0 / 0.0 min | Avg session: 230 min | Truncated: 5 | Failed patient: 5

[FAIL]" if failed else "[PASS] Strategy: FIXED | Wait (mean / max): 0.0 / 0.0 min | Avg session: 230 min | Truncated: 5 | Failed patient: 5